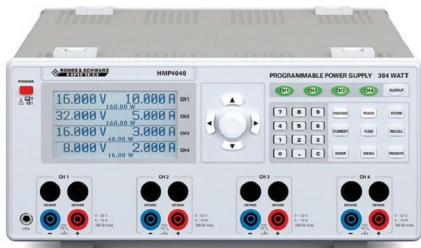


Some Multi-Channel Programmable DC Power Supplies with Built-In PFC

Brand	Model number	Voltage range	Maximum current	Maximum output power	Programming accuracy		Interface	Price
					Voltage	Current		
Keysight	E3631A	-25V to 25V	5A	80W	Voltage 0.05% + 20mV, 0.05% + 20mV, 0.1% + 5mV	Current 0.15% + 4mA, 0.15% + 4mA, 0.2% + 10mA	GPIO and RS232 standard	US\$ 1386.00
National Instruments	NI PXI-4110	-20V to 20V	1A	9W (from PXI backplane) or 46W (from APS-4100)	$\pm 0.05\%$ + 4mV, ± 0.05 + 10mV, ± 0.05 + 10mV	± 0.15 + 4mA, ± 0.15 + 60 μ A, ± 0.15 + 4mA		US\$ 1687.95
Rigol	DP831A	-30V to 30V	5A	160W	0.1% + 20mV	0.2% + 10mA	USB host & device, LAN, RS232, digital I/O, USB-GPIB (optional)	US\$ 790
Rohde & Schwarz	HMP4040	0V to 32V	10A	384W	<0.05% + 5mV (typ. ± 2 mV)	<0.1% + 5mA (typ. ± 1 mA at I < 500mA)	-Dual Ethernet/USB interface -IEEE488 (GPIB) interface	US\$ 2227.75



Rohde & Schwarz HMP4040 programmable four-channel power supply

converter that ensures complete security and exceptional stability to control settings.

EMC-safe. Another feature of the new PPS is its safe design that meets electromagnetic compatibility (EMC) standards. This EMC-safe PPS comes with a fully-active PFC that reduces the level of harmonic current that is taken from AC supply lines. The so-designed PPS also allows over-voltage trip adjustments for all outputs.

Extendable to DC and AC + DC mode. This feature added to the PPS provides a DC voltage component as per requirement. It provides an external analogue input to amplify the scalable analogue signal from the available arbitrary signal generator. It has a self-diagnosis feature, which ensures reliability and quality in testing automatic test equipment.

Noise detection. Some of the new PPSs are able to measure frequency harmonics. Dheeraj Kapoor, assistant business manager, Aimil Ltd says, "Frequency harmonics in the PPS helps measure the flowing current or noise in the application it is being used in. The amount of noise so measured can be easily identified by test engineers, and respective resolutions for the same can be worked on." ●

CONTRIBUTORS TO THE STORY

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The mode indication informs if the output is getting generated in constant current or constant voltage mode, whereas the automatic crossover allows the change of mode automatically, based on requirement. These also have fine controls that allow the output voltage to be set within 10mV, whereas to limit the current, the control is loga-

rithmic so as to give good resolutions at low-current settings.

Internal lock. New PPSs have internal lock buttons that transfer the control of current and voltage to internal digital circuits by simply pressing a switch. These are initially controlled by analogue controls. This lock is kind of a digital-to-analogue